

A survey of artificial intelligence methods for renewable energy forecasting: Methodologies and insights

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ABSTRACT

The efforts to revolutionize electric power generation and produce clean and sustainable electricity have led to the exploration of renewable energy systems (RES). This form of energy is replenished and cost-effective in terms of production and maintenance. However, RES, such as solar and wind energies, is intermittent; this is one of the drawbacks of its usage. In order to overcome this limitation, studies have been undertaken to forecast its availability and power output. The current trending method of forecasting availability and the power generated by the RES is the artificial intelligence (AI) method. However, with all its potential, traditional AI, such as Artificial Neural Network (ANN), Support Vector Machine (SVM) and many more, does not have it all. Because of this, metaheuristic algorithms are being explored as optimization techniques to increase the performance accuracy of these AI forecasting methods and overcome some challenges of traditional AI models. This study presents an insightful survey of AI (traditional AI and metaheuristic) methods explored to forecast the availability of solar and wind renewable energy systems. A survey of the existing surveyed literature was presented. The taxonomy of the explored AI in RES was formulated, and the theoretical backgrounds of some of the traditional AI algorithms were presented. Also, the various forms of metaheuristic algorithms and the improved versions applied to optimize classical AI methods to forecast RES systems' availability and power output were surveyed. A conceptual framework of the hybrid AI application in RES was formulated. Finally, the survey discussion, insight, challenges of the existing models and future directions were presented.

1. Introduction

Since its invention, electrical energy has contributed to society's development and sustainability. Industrial and economic growth directly relates to the supply of electrical energy. A sustainable power supply contributes to the standard of living of every citizen globally. The forms of generating electrical energy are thermal power generation and renewable power generation. In thermal power generation, thermal energy is used to produce electrical energy. The raw materials for thermal power generation systems are nuclear, coal, gas and oil plants [1]. Nuclear power plants have been at the forefront of power generation. However, building and maintaining a thermal power plant is capital-intensive. The negative impact of thermal power plants on society cannot be overemphasized. Emitting pollutants like carbon-related substances from fossil fuels depletes the citizens' life expectancy [2]. The by-products of these plants have increased global warming and

climate change through greenhouse emissions [3,4]. Due to all these negative impacts of thermal power plants on society, the preference for renewable energy systems (RES) has increased globally.

The advantages of RES over thermal generation systems are enormous and, at the same time, cannot be underestimated. The reduction of greenhouse emissions, low global climatic change [5], and low cost of production and maintenance are parts of its advantages [6]. Nonetheless, RES, such as solar and wind, are challenged by their unstable nature and poor estimation and optimization, making them difficult to integrate into a national grid system [7,8]. Many models, like persistence, physical (Numerical weather prediction, NWP), regression, and statistics, have been deployed to mitigate the stochastic nature of renewable energies [9]. Each of these models has one shortcoming or the other, which necessitates the development of the other. For instance, the persistence model performs well in short-term forecasting, while the physical model is perfect for medium-term or long-term forecasting [10]. In order to ensure optimal planning, estimation, integration and

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Nomenclature*Acronyms*

ABC	Artificial Bee Colony	HBO	Heap-Based Optimization
AC	Alternating Current	HC	Hierarchical Clustering
ACO	Ant Colony Optimization	HNN	Hermite Neural Network
AHA	Artificial Hummingbird Algorithm	HOMER	Hybrid optimization model for electric renewables
AI	Artificial Intelligence	HY-BRID2	Hybrid Power System Simulation Model
ANFIS	Adaptive Neuro-Fuzzy Inference System	IAE	Integral Absolute Error
ANN	Artificial Neural Networks	IVMD	Variational Mode Decomposition
AOAHA	Adaptive Opposition-AHA	KNN	K-Nearest Neighbor
ARIMA	Autoregressive Integrated Moving Average	LP	Linear Programming
ARMA	Autoregression Moving Average	LR	Linear Regression
BBO	Biogeographical-Based Optimization	LS-SVR	Least-Square Support Vector Regression
BDA	Big Data Analysis	LSTM	Long Short-Term Memory
BER	Al-Biruni Earth Radius	MA	Memetic Algorithm
BFO	Bacteria Foraging Optimization	MAE	Mean Absolute Error
BPNN	Backpropagation Neural Networks	MAPE	Mean Absolute Percentage Error
CA	Cultural Algorithm	MCC	Maximum Correntropy Criterion
CM	Mixture Correntropy	MFO	Mayfly Optimization
CNN	Convolutional Neural Network	MIP	Mixed Integer Programming
CRO-SL	Coral Reefs Optimization with Substrate Layer algorithm	ML	Machine Learning
CSA	Cuckoo Search Algorithm	MLP	Multilayer Perceptron
DBN	Deep Belief Networks	MLR	Multi-Linear Regression
DC	Direct Current	MMF	Multi-Model Framework
DE	Differential Evolution	MTER	Mean Training Error Rate
DF	Dragonfly Algorithm	NAR	Nonlinear Autoregressive
DNA	Deoxyribo Nuclear Acid	NARX	Nonlinear Autoregressive Exogenous
DNN	Dynamic Neural Networks	NCPR	Nonlinear Consequent Part Recurrent
DT	Decision Tree	NIO	Nonlinear Input-Output
EA	Evolutionary Algorithm	PBF	Polynomial Basis Function
EDA	Estimation of Distribution Algorithm	PBIL	Population-Based Incremental Learning
EFO	Electromagnetic Field Optimization	PSO	Particle Swarm Optimization
EL	Ensemble Learning	PV	Photovoltaic
ELM	Extreme Learning Machine	R	Coefficient of Correlation
EMD	Empirical Mode Decomposition	RBF	Radial Basis Function
EMP	Extended Mathematical Programming	ReLU	Rectified Linear Unit
ENN	Elman Neural Networks	RE	Renewable Energy System
ES	Evolutionary Strategy	RF	Random Forest
ETR	Extra Tree Regressor	RMSE	Relative Mean Square Error
FAC	First-order Adaptive Coefficient	RNN	Recurrent Neural Networks
FFA	Firefly Algorithm	SAC	Second-order Adaptive Coefficient
FL	Fuzzy Logic	SAMME	Stagewise Additive Modeling
FTMA	Fine-Tuning Metaheuristic Algorithm	SAMME.R	Real SAMME
GA	Genetic Algorithm	SARIMA	Seasonal Autoregressive Integrated Moving Average
GBDT	Gradient Boosting Decision Tree	SCE	Shuffled Complex Evolution
GDHS	Global Dynamic Harmony Search	SE	Sample Entropy
GEP	Gene Expression Programming	SFLA	Shuffled Frog Leaping Algorithm
GP	Genetic Programming	SI	Swarm Intelligence
GPR	Gaussian process regression	SPSA	Sparrow Search Algorithm
GRACH	Generalized AutoRegressive Conditional Heteroskedasticity	SSA	Single Spectrum Analysis
GRNN	Generalized Regression Neural Networks	SVM	Support Vector Machine
GRU	Gated Recurrent Unit	TPA	Temporal Pattern Attention
GSA	Gravitational Search Algorithm	TSA	Tree-Seed Algorithm
GWO	Grey-Wolf-Optimizer	UV	Ultra-Violet
		WOA	Whale Optimization Algorithm
		WPT	Wavelet Packet Transform
		XGBoost	Extreme Gradient Boosting

operation of RES, scholars are taking giant steps to implement and address the challenges associated with the earlier-mentioned models. Some of the applications of AI in RES are forecasting the RES variability [11], maximum power point tracking of RES [12], and optimization of hybrid RES [13–15]. In the forecasting of the RES variability, AI methods such as ANN [13,14], SVM [16], LS-SVM [17], and fuzzy

inference systems [18] have been implemented. Recently, researchers investigated deep learning methods such as RNN, LSTM, and CNN [19] and [19,20]. Likewise, Ensemble Learning (EL) models such as Random Forest (RF) and CatBoost [21], Extreme Gradient Boosting (XGBoost) [22], and the ensemble of traditional ML [23] and Reinforcement Learning [24,25] have been implemented. The outcomes of the AI

implementations are assisting the practitioners in the field to make quality and informed decisions. Forecasting solar and wind systems is possible because of the meteorological and RES data availability. Predicting the unstable nature of solar and wind resources is being dominated by AI methods, and research has revealed that this trend will continue for decades.

2. Related study, motivation and contributions

Many studies published on the forecasting of solar and wind RES present the challenges and solutions with future opportunities in the field of RES. These studies provide potent mechanisms for planning, modelling, monitoring, diagnosing faults and fault adaptive operation of the electric power systems. In order to establish and summarize the existing published studies on AI implementation in RES, several review articles have been published for the benefit of researchers. The reviews pinpointed several AI methods applied to forecast the variability of solar and wind renewable systems: ML, DL, EL and hybrid methods. These studies emphasized the benefits and challenges of AI methods. Some of the benefits from the reviews are ML is less susceptible to noise and has less computational cost compared with the persistence and NWP models [26]. Also, it has been established that DL algorithms always result in better performance [27]. However, in those reviews, some of the challenges of AI in forecasting the variability of solar and wind renewable systems, such as data preprocessing and data clearing processes, were discussed [28]. They pointed out that machine learning can be, at times, biased, and the whole process can become useless if a suitable machine learning model is not applied [29]. In addition, some reviews discussed the metaheuristic method implemented to improve the performance accuracy of the AI methods. Nonetheless, with all the capabilities, metaheuristic methods increase the computation time of the forecasting models [11,30].

The existing review/survey articles either focus on one arm of the AI method or another. Some considered only ML, DL or metaheuristic methods, and some considered two arms, such as ML and metaheuristic methods. The most recent review that considered all the arms of the AI model for solar and wind renewable was published in 2021 [30]. This study's novelty is considering all the arms of the AI method applied to forecast solar and wind renewable systems. Table 1 presents the Comparative analysis of the existing review studies. This study will provide an accurate survey of the latest articles published on AI methods adopted to forecast the variability of global radiation of solar systems and the wind speed of wind systems. Also, the combination of hybrid AI forecasting models, which comprises traditional models and metaheuristic algorithms, and improved metaheuristic methodologies will be summarized. Findings are presented based on the existing research articles and future directions in forecasting solar and wind renewable energy. The contributions of this survey are:

- i. Comparative studies of traditional AI combined and hybridized AI systems with metaheuristic methods explored to forecast solar and wind RES are presented.
- ii. Little studies have been carried out on improving metaheuristic algorithms and their implementation on RES forecasting. This paper presents a few studies which improved metaheuristic algorithms and successfully applied them for RES forecasting.
- iii. Present future research directions that will help researchers and practitioners in solar and wind RES make informed decisions.

The remaining part of this survey article is organized as follows: Section "Related study, motivation and contributions" briefly discusses the forms of RES, the solar, the wind, and their hybrid forms, which is the focus of this survey. Section "Renewable energy" presents a taxonomy of AI in RES and discusses types of AI already applied for forecasting solar radiation and wind systems. Section "AI in RES" summarizes the metaheuristic algorithms employed for renewable

Table 1
Comparative analysis of the existing review studies

Author	Year	RES	Review Focus
[31]	2019	Solar and wind	DL: deep belief network, stack auto-encoder, deep RNN, data preprocessing, error post-correction, feature extraction and performance evaluation metrics
[32]	2020	Solar, wind, geothermal and electricity load demand	Time horizon, ML, EL, ANN, energy consumption for utility, private and government building
[11]	2020	Solar, wind, hydropower, biomass, wave, tidal and geothermal	ML, metaheuristic methods, data preprocessing, parameter selection algorithms and performance evaluation metrics.
[30]	2021	Solar and wind	ML: ANN, SVR, RNN and ELM, metaheuristic methods, target and time horizon, and model accuracy.
[26]	2021	Solar and wind	DL, dataset and data preprocessing, time horizon, optimization method and performance evaluation metrics
[33]	2021	Solar, wind and electric power load	Dataset, DL, load and energy generation in residential and commercial sectors
[34]	2022	Solar and wind	ML, hybrid spatiotemporal and vector autoregressive methods, decomposition methods, remote sensing and benchmark models
[29]	2021	Solar	ANN, BPNN, ANFIS and metaheuristic methods
[35]	2022	Solar, wind and electrical load	NWP, ML, DL, EL, spatial and temporal scale of dataset with performance metrics
[9]	2022	Solar and wind	Physical, statistical, ML and DL
[27]	2023	solar	ML, DL, metaheuristic methods, dataset, time horizon and performance metrics
[36]	2023	solar	The practicability of models such as regression, satellite-driven, statistical, AI and hybrid performance evaluation metrics of these models

energy forecasting. Section "Metaheuristic algorithms" provides an AI hybrid conceptual framework for renewable energy forecasting. Section "Hybrid AI in RES" presents findings and discussion based on the survey. Section "Conclusion" concludes with future research trends and recommendations.

3. Renewable energy

Harnessing natural resources like sun, wind, and water for human development and sustainability has received significant attention for over a decade. The explorations of these resources as RES have attracted studies in the electrical energy domain. The cost of building, running, and maintaining nuclear power plants, their environmental degradation, and the anxiety of exhausting the remaining fossil fuel in the globe contribute to the deployment of RES [1]. RES sources, such as hydropower, solar, wind, biomass, fuel cell, geothermal, and ocean energy, are frequently replenished and increasingly popular for generating clean power [37]. They significantly reduce fossil fuel consumption and substantially lower greenhouse emissions [2]. Mega-project investments in solar and wind energy have the potential to revolutionize the electrical power industry [38].

3.1. Wind source

The environmental benefits of wind energy sources, such as free air or water pollution, free toxic or hazardous substances, and free threat to public safety, have promoted it in the world of RES. Wind turbines accounted for 8.4 % of all utility-scale power output in the United States

in 2020. This number is projected to rise to 20 % in 2030 and 35 % in 2050 [39]. One of the challenges of wind energy generation is integrating wind speed into the power system. Accurate wind speed prediction is essential to integrate wind turbines into electrical systems.

The practical approaches for wind speed prediction are physics-based and data-driven methods [40]. The physics-based methods applied the equation of atmospheric motion to determine the development of meteorological observations. The method is characterized by computation complexity. Conventional time-series methods like the auto-regressive moving average (ARMA) and its variance have been applied to predict hourly wind speed [41,42]. The method performed better when the wind speed exhibited regular changes, but the ARMA and its variance performance degraded in the case of irregular wind speed variation. The current trend to overcome the shortcomings of the physics-based approach is the implementation of traditional AI methods and metaheuristic algorithms such as particle swarm optimization [43], genetic algorithm [44], and grey wolf optimization [45].

3.2. Solar source

Solar energy is the cleanest renewable energy source, and its adoption is exponential worldwide, both on small and large scales. According to the Renewable Energy Policy Network forecast for the 21st Century, total solar energy output is expected to exceed 8000 GWatt in 2050 [46]. Solar Energy usage is classified into passive solar and active solar. Passive solar energy makes direct and indirect use of photovoltaic (PV) energy from the sun. Direct use converts the DC energy from PV to AC form for building or structure use. Due to variations in the intensity of light, the energy from this form can fluctuate. The indirect form is concentrated solar power, which uses mirrors or lenses to produce heat to power a steam turbine that generates an electric current. Active solar energy uses the sun's electromagnetic radiation to generate electrical power. Silicon Boron semiconductor solar chips are primarily used. The problem with these chips is that they have a low-efficiency ratio and can only be used to supply the energy needs of small devices [1]. The elements that affect solar energy output demand investigation before any renewable energy production facility can be designed. Factors that affect solar radiation are well-understood and are being considered in every part of the world.

3.3. Hybrid RES

The seasonal intermittent and fluctuating renewable energy sources significantly affect the RES's power output. Dependency on one source cannot meet the energy demand. Research and technological advancements have compounded two or more renewable energy sources to form hybrid RES [47,48]. The most critical issue in hybrid RES is the optimization and sizing of the system. Software is being developed to optimize and size the hybrid RES optimally. These include the Hybrid Optimization Model of Electric Renewable (HOMER) [49], the Hybrid Power System Simulation Model (HY-BRID2), and the Hybrid Optimization using Genetic Algorithm (HOGA) [50]. Other forms of RES are biomass, hydrogen energy, and geothermal energy. Some of these other forms have been hybridized with wind and solar RES to enhance the production sustainability of electrical power generation. In the quest to enhance the power output of RES, the geothermal organic Rankine cycle was combined with a solar energy system [51]. In Ref. [52], a hybrid system comprising solar, wind, hydrogen, and diesel generators was fused to study dynamic multi-objective strategy. In Ref. [53], solar photovoltaic system and biomass were combined to overcome the intermittent nature of RES. Several studies were conducted on these forms for RES, but this survey mainly focuses on solar and wind RES.

4. AI in RES

The field of computer science and engineering that creates a smart

and intelligent system that can behave like a human and make a decision is known as artificial intelligence. It explores the understanding of human thinking to develop clever entities that perform efficiently for some complex problems [29]. This study of understanding human thinking is still ongoing due to the complexity of the human brain. The success achieved so far in the AI domain has reduced the problem of manual complex computation. AI applications have cut across many fields, such as information retrieval, database, production, planning, accounting, distribution economy, product design, industry, electrical power, medicine, security and forensics, and food quality monitoring. Likewise, in the application of RES, AI has been applied to forecasting and estimation [54], optimal sizing [55,56] and operation strategy [57,58]. Fig. 1 presents the taxonomy of the different AI methods to predict RES. Moreover, Table 2 presents the traditional methods of AI implemented to forecast the solar and wind RES. The AI theory is based on statistical, neural, and heuristic learning.

4.1. Artificial neural network (ANN)

Neural learning is the most applied learning in AI, and ANN is the central technique of neural learning. In 1943, McCulloch and Pitts proposed the hypothesis of the mathematical model for a primitive cell (neuron) of the human brain to establish the theory of ANN [70]. The neuron is triggered when the weighted sum of input exceeds a threshold value, and the activation function is applied. After that, an output response is produced. ANN is a robust learning method because it can fine-tune its values and fix its output error. Some of the variances of ANNs are the Feedforward neural networks, Radial basis function neural networks (RBFNN), and Kohonen self-organizing network [71]. Generalization, parallel processing, adaptability and handling of nonlinear systems are some of the strengths of the ANN method [72]. An ANN-LM model was applied in Ref. [64] to conduct hourly short-term solar radiation forecasting. A site dataset was utilized in the study, and the min-max normalization method was applied to preprocess the dataset. The study in the Ref. [60], an MLP-NN was applied to predict daily solar energy using the Netherlands meteorological dataset. Rrelieff and min-max were applied as preprocessing methods. The performance metrics MAE and RMSE depicted the capacity of the model. MLP-NN with Rrelieff and min-max outperformed all the benchmarked models, as revealed in Table 2. In another study, the advantages of ANN were explored to conduct hourly short-term forecast. This study differs from the earlier The model exhibited computation complexity. In another study, ANN model was explored to conduct hourly short-term forecasts of PV power [68]. This study differs from the earlier mentioned study because it took the advantage of ultraviolet index of the Alice Spring Australia dataset. The performance analysis proved the advantages of the study. Nonetheless, black-box nature, overfitting, and complex hyperparameter tuning are some of the drawbacks of these ANN models [73,74].

4.2. Support vector machine (SVM)

The SVM algorithm belongs to the categories of soft computing methods applied for classification and regression problems. It employs a linear model to implement nonlinear class boundaries and can be applied to linear and nonlinear problems. The method transforms the original training data into a higher dimension using nonlinear mapping [75]. Then, it searches for the linear optimal separating hyperplane with the new dimension. The linear optimal separating hyperplane is a decision boundary separating the tuples of one class from another [76]. Therefore, the basic concepts of SVM are separating hyperplane, maximum-margin hyperplane, soft margin, and kernel function. A hyperplane is a linear decision surface that divides the space into two parts [63]. The benefits of SVM are flexible kernel selection, outliers' robustness, less over-fitting and a compact description of the learned model [65]. However, the SVM algorithm has shortcomings of

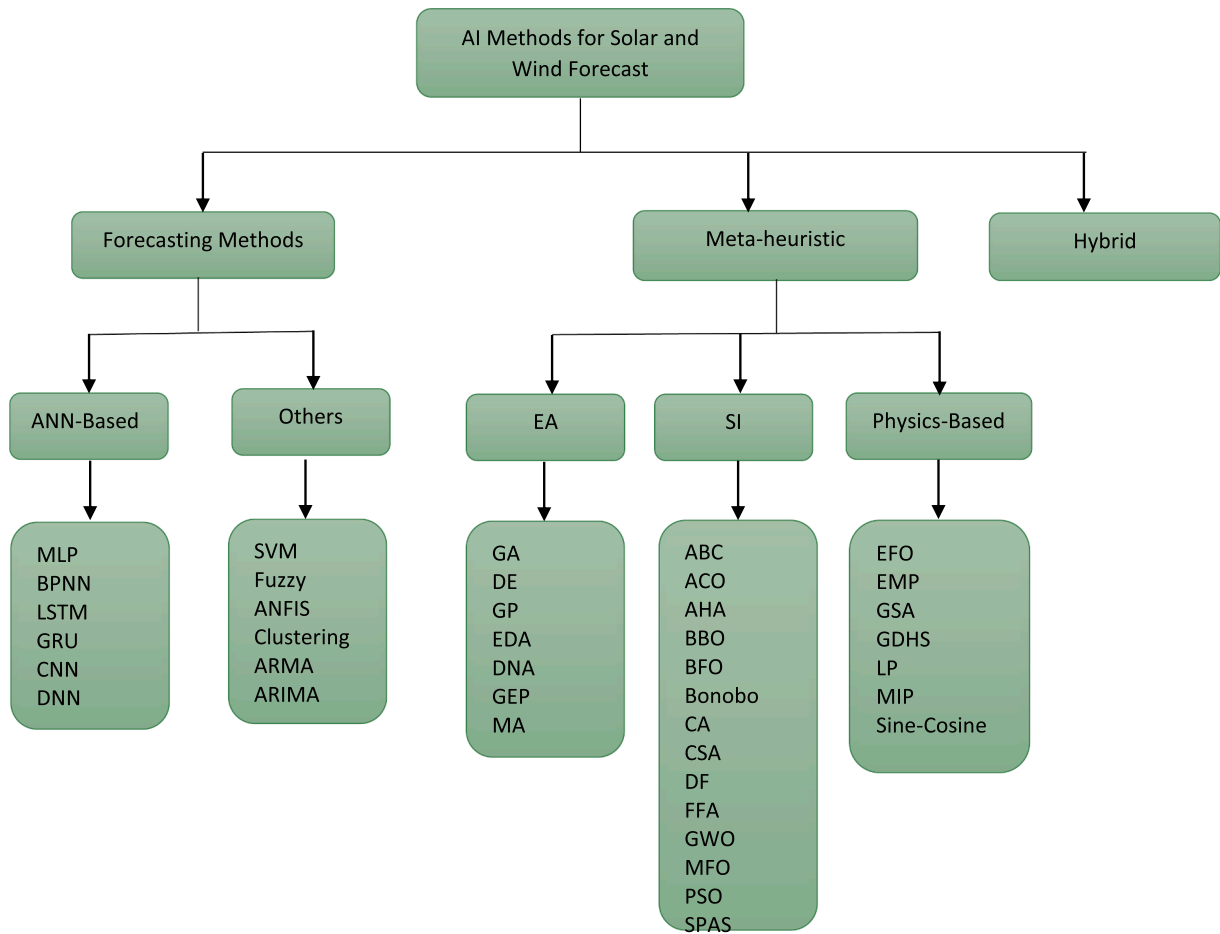


Fig. 1. Taxonomy of AI Forecasting Methods of Solar and Wind RES

inappropriate hyperparameter selection and handling of the large dataset that reduces the model's performance [77,78].

4.3. Long short-term memory (LSTM)

ANN models have evolved from one variant to another; Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN) and Deep Neural Networks (DNN) are some of the variants. The input vectors of these variants are independent of each other. Unlike CNN and DNN, RNN produces sequential information from time series data, uses the hidden state to maintain past information and regularly updates new ones [79]. It has a broad scope of applications in dynamic system predictions. In some applications, the output of RNN was fed back as input to constitute a well-described dynamic system [80]. It has a unique deep structure and performs well in time series applications.

However, the weakness of RNN is the complex training algorithms explored during tuning. This process usually results in vanishing or exploding gradient problems, making it challenging to learn the long-term dependencies [81]. In order to address this weakness of RNN, LSTM and GRU were proposed [82,83]. These two algorithms are variants of RNN capable of learning long-term dependencies to generate sequential information from time series data. LSTM has a memory cell (hidden layer) that stores information and recursively computes new carry with a gating mechanism. These are the forget gate, update gate and output gate [84,85]. The most significant element of LSTM operation is cell state c .

Consider the LSTM architecture in Fig. 2; the input set is given as $x_k \in \mathcal{R}^n$, n represents the number of the set. The forget gate f_k determines the information needed to be discarded from the cell state based on the

previous operation h_{k-1} and is formulated as equation (1).

$$f_k = \sigma(W_f * [h_{k-1}, x_k] + b_f) \quad (1)$$

where σ is the sigmoid nonlinear activation function and b_f represents the bias function of the forget gate.

The second gate in the LSTM block is the update gate p_k , which determines the amount of information in the memory cell. It comprises the input gate i_k and the candidate carry $\hat{c}$. The input gate i_k defines the memory state values to be updated while the candidate carry, $\hat{c}$, defines the candidate value that will update the memory state. The expressions for these three components are given as the equations (2)–(4).

$$i_k = \sigma(W_i * [h_{k-1}, x_k] + b_i) \quad (2)$$

$$\hat{c}_k = \tanh(W_c * [h_{k-1}, x_k] + b_c) \quad (3)$$

$$p_k = i_k * \hat{c}_k \quad (4)$$

where b_i and b_c represent the bias functions of the input gate, and candidate carry, $\tanh$ and hyperbolic tangent functions.

Now, the updated memory state c_k is expressed as equation (5)

$$c_k = p_k + f_k * c_{k-1} \quad (5)$$

The third gate of LSTM consists of the output gate o_k and the LSTM block output layer h_k . Moreover, the output gate o_k is formulated as equation (6).

$$o_k = \sigma(W_o * [h_{k-1}, x_k] + b_o) \quad (6)$$

where b_o represents the bias functions of the output gate function.

Table 2
AI methods for RES

Reference	AI Model	Forecasting Horizon	Data Site	Preprocessing Method	Application Area	Comparison Algorithm	Performance Metric	Findings
[59]	Deep CNN	Hourly	Taiwan	-	Solar Radiation	SVM, RF, DT, MLP, LSTM	MAE	Good generalization and robustness in providing high accuracy
[47]	Ensemble MLP Regressor,	Hourly	Turkey	RFECV	Solar and Wind Systems	AdaBoost, SVR, KNR, GPR, LR, ETR	R ² MSE MAE Computation Time	The model provides significant statistical learning; nonetheless, implementing an optimization model addresses the premature convergence of the model.
[60]	MLP-NN	Daily	Netherland	RreliefF, Min-Max	Solar power forecasting	Persistence, Smart Persistence, SVR, RF, NN Ensemble	MAE MRE RMSE	Data preprocessing and feature extraction with RReliefF enhanced the performance accuracy. However, the implementation of the optimization method will improve the search space for the global minimum.
[18]	SVR-RBF	Daily and Monthly	Iran	-	Solar Radiation	SVR-PBF, PSO-SVR	R ² MAPE MAE RMSE RRMSE	Data preprocessing and enhanced optimization methods can improve the proposed model's performance.
[61]	SVR-RBF, PBF	Hourly	Zahedan and Bojnurd, Iran	-	Solar Radiation	Empirical Model	R ² RMSE	The method outperformed the compared models. Data preprocessing and tuning of SVR parameters will reduce performance errors.
[62]	Time Delay NN MMF	Hourly	NTU, China	K-means clustering with dynamic time-warping	Solar Radiation	ARMA	Symmetric MAPE	Data segmentation using the K-means algorithm with dynamic time warping boosts the prediction model's performance. Meanwhile, increasing the clustering accuracy increases the prediction model's performance.
[63]	Fuzzy-SVM	Daily	Turkey	Clustering with FCM	Solar Radiation	ANFIS, Kernel Function	RMSE MAE MaxAE IQR-AE	Implementation of the data preprocessing and optimization algorithm will overcome the challenge of easy convergence of the proposed model.
[64]	ANN-LM	Hourly	FAO-UN India	Min-Max	Solar Radiation	-	R MSE	The proposed strategy exhibited computation complexity and local minimum. Applying data cleaning, preprocessing, and metaheuristic algorithm will improve the proposed accuracy.
[65]	SVM	5mins	NREL Jeddah, Qussim Saudi Arabia	-	Solar Radiation with Tilt Analysis	ANN	R ² RMSE MRE	Implementation of data preprocessing and optimization methods will address the forecasting model's premature convergence and local minimum shortcomings.
[66]	LSTM	Monthly	CESM	-	Wind power forecasting	-	MAE	Data preprocessing and implementation of an improved optimization will impact the forecasting capacity of the proposed model.
[67]	Dual-Stage Attention LSTM	Daily	North China	VMD	Wind power forecasting	XGBoost, MLP, LSTM, CNN-LSTM, TPA-LSTM, Transformer	MAE RMSE R	Implementation of data decomposition and metaheuristic algorithm will result in better performance
[68]	ANN	Hourly	Alice Spring Australia	-	PV Power Forecasting	-	MSE MAE Accuracy	Investigation of the UV index contributed to the PV power forecasting accuracy; nonetheless, implementing metaheuristic methods will improve the model's superiority.
[69]	LSTM	Hourly	Ecuador Tungurahua	Correlatiom matrix, Normalization, structuring and conversion	Wind speed Forecasting	Persistence, MA, ARMA, NARX, NAR, NIO	R MSE MAPE	The model requires an optimization model to address the challenges of premature convergence and local minimum.

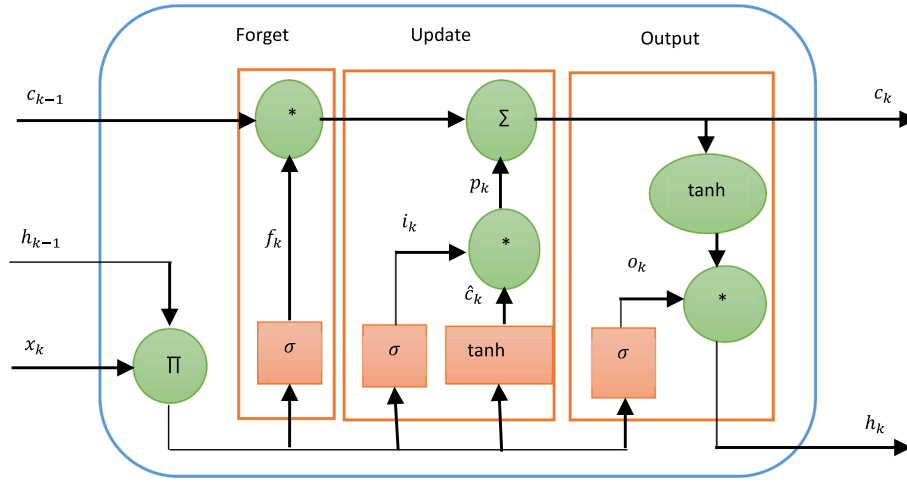


Fig. 2. LSTM Block Layer Structure

o_k is utilized to update the desired function of h_k , and it is expressed as equation (7)

$$h_k = o_k * \tanh(c_k) \quad (7)$$

The parameters W_f , W_i , W_c and W_o denote the trainable weight vectors of the LSTM block layers. The implementation of the LSTM neural network has the potential to handle learning information over extended periods through recurrent backpropagation, which takes a long time.

4.4. Ensemble learning (EL) model

Ensemble techniques are well-described as ML methods that use multiple base learners called weak learners to unravel problems. ELs are different from conventional ML methods because they use one dataset to create more than one theory and then merge them at the end of the procedure [87]. Therefore, the advantage of ensemble methods is that they explore more than one classifier to improve the prediction model. Ensemble methods are most utilized because they are more robust and accurate than traditional ML methods [88]. The EL methods in ML are generally categorized into three main sections: bagging, boosting, and stacking. The blending method and super ensemble are examples of stacking methods [89]. One of the advantages of EL methods is that it reduces the generalization error of the base learners.

4.4.1. Bagging EL method

Breiman developed the bagging ensemble method, also known as bootstrap aggregating, in 1994 to improve the performance of ML models [90]. In the bagging method, the classifier builds a model for each sample after generating random samples. Each sample's final results are merged using majority voting [88]. Bagging EL results in diversity by producing bootstrapped duplicates of the features, where various input data subsets are selected randomly with substitutes from the earliest training dataset. The bagging method includes Random Forest (RF), Random Patches (RP) and an Extra Tree classifier [91]. The benefits of bagging are as follows:

- It diminishes variance without improving the bias and input data diversity due to the bootstrap technique.
- Bagging can manage the effect of missing values in the dataset.
- The algorithm enhances the base learner performance in the case of an unstable learner algorithm.
- The bagging trains model with lesser sample data; therefore, it has less computational cost than other ML methods.

However, the drawback of bagging EL is interpretability difficulty

because it utilizes many decision tree algorithms [92]. Bagging EL variances are QuasiBagging, DoubleBagging and OverBagging [93].

4.4.2. Boosting EL method

The boosting EL method is an ML approach that converts a collection of weak learners to a robust classifier. This method was first proposed in Schapir 1990 to produce a single robust learner from a set of weak learning algorithms [94]. The fundamental clue of boosting is to iteratively engage the base learning algorithms to the modified versions of the input dataset. Boosting procedures explore the input dataset to train, predict and select misclassified training samples from a weak learner. The successive weak learner is trained with a modified learning dataset containing the misclassified cases resulting from the former learning set. The progression maintains these iterations until the set figure of base learners is achieved and a weighted operation is performed on the base learners. The inaccuracy is calculated, and the weight of the accurately classified sample set is reduced. The misclassified sample set is given additional weight, triggering the base learner to give more concentration to the sample set. The ultimate model achieved by the boosting method is a linear combination of various base classifiers weighted for their excellent quality performance [95]. When a base learning algorithm is biased alongside particular cases, those cases are granted further weight; then, the learning algorithm will correct the bias. AdaBoost, gradient boosting, and XGBoost are variances boosting algorithms [96,97]. The significant advantage of boosting is that it can handle base learning algorithms with low variance and high bias to produce a predictive model with low variance and low bias. The main focus of boosting is to achieve better accuracy from weak learners.

Nevertheless, boosting does not thrive well in an unstructured dataset because of the iterative learning technique, which may lead to overfitting. The weight given to the noisy dataset will be bigger than the weight given to the other sampling cases, pressing the predictive algorithm to concentrate significantly on the unstructured sampling cases [87,88].

4.4.2.1. Adaptive Boosting (AdaBoost). AdaBoost is the most exemplary boosting method widely applied as an ensemble learning technique. A base learner is trained in every iteration of AdaBoost. The weights of the training sample are updated based on the performance of the earlier iterations. Several variants of AdaBoost exist, which are determined based on how weights are updated and which samples are affected. The first AdaBoost algorithm was created to solve binary classification problems. The algorithm's base learners predicted the probability of the target class [98]. This method involved the relative adjustment of the weight w_i of each instance x_i to the probability of correctly predicted

samples and inversely related to the error E_t of the classifier T_t . Moreover, the classifier's decision on the final prediction of a new sample is also weighted based on its accuracy during the training period [91]. Another variant of AdaBoost is a multi-class variant, namely AdaBoost.M1 [98]. In AdaBoost.M1, only the weights of the correctly predicted instances are reduced. This weight reduction is still carried out for the error produced by the base learner. Each classifier prediction is reweighted by its accuracy.

For these two variants to perform correctly, the error of each base learner is assumed to be less than 0.5. If not, the correctly predicted samples will be the ones that will be boosted instead of the misclassified samples. Additionally, the contribution of each classifier to the final decision becomes negative. Other versions of AdaBoost are AdaBoost.M2, AdaBoostMH, MultiBoost and Stagewise Additive Modeling (SAMME) and Real SAMME (SAMME.R) [91]. All these versions are the extension of AdaBoost to solve the multi-class problem and obtain better performance accuracy.

Considering a q labelled training instances $S = \{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_q, y_q)\}$, where y_i represents the target label of the input feature x_i , the exact weight D_1 of the input feature x_i and the exact weight update D_{t+1} are determined using equations (8) and (9)

$$D_{1(i)} = \frac{1}{n}, \quad i = 1, 2, \dots, q \quad (8)$$

$$D_{t+1(i)} = \frac{D_{t(i)} e^{(-\alpha_t y_i h_t(x_i))}}{z_t}, \quad i = 1, \dots, q \quad (9)$$

where $h_t(x)$ denotes base learner, $t = 1, \dots, T$ denotes iterations number, z_t denotes factor normalization, while α_t represents learning algorithm weight $h_t(x)$ when obtaining the final classifier prediction. When the AdaBoost algorithm is being implemented, the incorrectly predicted cases in $h_t(x)$ are allotted bigger weights in the $t+1$ training cycle. Moreover, z_t is chosen that D_{t+1} will generate a dispersal sample. z_t plus α_t are determined with equations (10) and (11).

$$z_t = \sum_{i=1}^n D_{t(i)} e^{(-\alpha_t y_i h_t(x_i))} \quad (10)$$

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right) \quad (11)$$

where ϵ_t is the error rate of the learner, and it is determined with equation (12).

$$\epsilon_t = P[h_t(x) \neq y_i] = \sum_{i=1}^n D_t(i) I[h_t(x_i) \neq y_i] \quad (12)$$

At the end of the number of given iterations, equation (13) determines the final classifier output.

$$H_{(x)} = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right) \quad (13)$$

AdaBoost uses decision trees as base learners to train data iteratively. Moreover, any other ML algorithm can be used as a base learner based on the application area suitability. It is flexible and increases the performance of the training set. Nevertheless, it resulted in overfitting when the dataset had noises and outliers.

4.4.2.2. Gradient boosting. Another variant of boosting EL technique is gradient boosting, which was introduced to optimize the performance of base classifiers [99]. This variance is also known as gradient boosting decision tree (GBDT) because it primarily employs decision trees algorithm as base classifiers. The fundamental principle of this variant of boosting algorithms is based on the development of base classifiers, which are exceedingly interrelated to the negative gradient of the cost

function [100].

Consider a dataset $S = \{x_i, y_i\}$, composed of independent variables and dependent variables. $i \in \{0, \dots, j\}$, j , represents the overall number of samples of the dataset. The gradient boosting method determines the approximation function, $\hat{F}_{(x)}$, that maps independent features x to the dependent features y through minimalizing the cost function, L . Therefore, gradient boosting utilizing the weighted sum functions to produce an additive estimate of $F_{(x)}$ is formulated as equation (14).

$$F_m(x) = F_{m-1}(x) + \rho_m h_m(x) \quad (14)$$

where ρ_m means the weight of m th formulation of $h_m(x)$. The continuous approximation of $F_{(x)}$ is expressed as equation (15).

$$F_0(x) = \underset{\alpha}{\text{argmin}} \sum_{i=1}^N L(y_i, \alpha) \quad (15)$$

Equation (16) minimizes the objective of base learners.

$$\rho_m h_m(x) = \underset{\rho, h}{\text{argmin}} \sum_{i=1}^N L(y_i, F_{m-1}(x_i) + \rho h(x_i)) \quad (16)$$

In equation (17), a greedy step approach is adopted using gradient descent optimization for F^* , each h_m is trained using a new dataset $D = \{x_i, r_m\}$ to generate with the first residuals rm_i .

$$rm_i = \left[\frac{\partial L(y_i, (x))}{\partial F(x)} \right] F(x) = F_{m-1}(x) \quad (17)$$

After that, ρ_m is determined using linear search optimization in a condition where h_m equal to a false residual, such as the quadratic loss function, the false residual equals zero, and the iteration stops. Moreover, hyperparameter regularization of this boosting method can be carried out with shrinkage to limit every gradient descent step, $F_m(x) = F_{m-1}(x) + l \rho_m h_m(x)$, where parameter l is the learning rate, it controls each base classifier by shrinking its contribution within $0 < l \leq 1$ [101]. The gradient boosting method can learn the complete structure from the input set and correct the errors of previous models. However, the algorithm can be overfitting if the iteration process is not adequately regularized. Also, it is not suitable for a noisy dataset.

4.4.2.3. Extremely Gradient Boosting (XGBoost). Chen & Guestrin, 2016 first proposed that the XGBoost ensemble method is highly scalable and can handle many machine-learning challenges [102]. It employs the gradient-boosting principle to lessen decision tree model errors and produces high-performance models. It is mainly used for classification and regression applications [100]. In XGBoost, overfitting is prevented because of the loss function with regularization terms. Consider a dataset $S = \{x_i, y_i\}$, composed of independent variables and dependent variables. $i \in \{0, \dots, j\}$, j , represent the total number of samples of the dataset as described in the gradient boosting method. The total number of decision trees in the model is D and the predicted estimates, $\bar{y}$ at a sample, i can be determined as equation (18)

$$\hat{y} = \sum_p^D f_p x_i \quad (18)$$

where $f_p x_i$ is the predicted count up of the instance i for the p th tree.

In the XGBoosting algorithm, an objective function is formulated to optimize the loss function using the outcome of the previous base learner and add a regularization technique to enhance its performance. The mathematical formulation of the initial objective function, the loss function and the regularization term are given in the equations (19), (20), and (21), respectively.

$$J = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{p=1}^D \Omega(f_p) \quad (19)$$

$$l(y_i, \hat{y}_i) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (20)$$

$$\Omega(f) = \zeta T + \frac{1}{2} \gamma \sum_{b=1}^T (X_b^2) \quad (21)$$

where n represents the number of instances processed at the p th trees and $\hat{y}_i$ represents the predicted output. T , b , X , denote the overall number of leaf nodes in the regression decision tree, the identification index of each leaf in a node, and the weight of a specific leaf node, respectively. The complexity constraint, ζ , regulates the least loss decrease gain mandatory for excruciating an internal node. and γ denotes customization parameters. Studies have proved that assigning high values to ζ and γ results in a straightforward decision tree structure and reduced risk of overfitting [102].

Applying a second-order Taylor Approximation method to the objective function J , the optimized version is given as the equation (22)

$$J^* = \sum_{i=1}^n \left[g f_m(x_i) + \frac{1}{2} h f_m^2 \right] + \Omega(h_m) \quad (22)$$

The g_i represents the first derivative while h_i represents the second derivative of the cost function. The final cost function in the sample S_j is determined by totalling the degree of the loss of all the leaf nodes j , then the objective function is formulated as equation (23)

$$J^F = \sum_{j=1}^T \left[\left(\sum_{i \in S_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in S_j} h_i + \gamma \right) w_j^2 \right] + \zeta T \quad (23)$$

where w is the weight of the tree leaves.

XGBoost method has many advantages over some of the conventional ML methods. It can handle overfitting by integrating a regularization term during the modelling process. Also, it is flexible and efficient due to its optimal management of processing memory resources [103]. XGBoost can handle missing values of the input set during the training procedure. Moreover, parallel computing of feature engineering is another strength of the XGBoost ensemble learning method [91]. Nonetheless, it produces less accurate results when it is used to train an imbalanced dataset. Another drawback of the boosting method is its high number of hyperparameters, making the algorithm tuning difficult.

4.4.3. Stacking EL method

Stacking is an EL technique that trains ML algorithms to combine predictions from multiple algorithms. The method was proposed to reduce the generalization error related to ML challenges [104]. The fundamental concept of stacking is applying the predictions of base learners as input metadata to another learner, known as the meta-learner, to train the predicted metadata. The predictions are used as features for training the meta-classifiers to produce the final prediction results. A meta-classifier is any conventional ML algorithm that best combines base learners' predictions. Base learner algorithms are described as level 0 learners (multiple ML-based algorithms), while the meta learners are described as level 1 learners [105]. The meta-learners stacked on the based learners are described as stacking.

The stacking framework has three significant steps: when the chosen algorithms train the input dataset. The second is the creation of a new data set using the predicted output of the base learners. The expected target forms the new feature, while the initial target variable remains the new target variable for the new dataset. The third step is training the meta-classifier with the newly generated dataset [106]. Linear classifiers like logistic regression algorithms are commonly employed as meta-classifiers [107]. Considering a training dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, equation (24) expresses the prediction of the base learners:

$$h_i = L(D_i) \quad (24)$$

Then, equation (25) represents the new dataset constructed as $\{\hat{x}_i, y_i\}$;

$$\hat{x}_i \{h_1(x_1), h_2(x_2), \dots, h_T(x_i)\} \quad (25)$$

The final prediction model is obtained by applying a meta-classifier $\hat{h}$ to train the newly constructed dataset as shown in equation (26).

$$H_{(x)} = \hat{h}(h_1(x), h_2(x), \dots, h_T(x)) \quad (26)$$

Part of the significant benefits of stacking is the exploration of generalization and diversity of different base learners using the same input dataset. High accuracy is also one of the advantages of the stacking method [108]. Nonetheless, a large stacking dataset can lead to high computational costs because the entire data set trains every base learner.

5. Metaheuristic algorithms

Just like neural learning, metaheuristic learning-based methods have been applied in different fields of applications. Metaheuristic algorithms are part of AI algorithms and are divided into three categories. These are evolutionary algorithm (EA), swarm intelligence optimization (SIO), and physics-based optimization algorithms. EA is a category of metaheuristic algorithms that employ deterministic algorithms and heuristic random search to acquire the nearly immediate solution to the optimization problem via iterative iterations. EA simulates the biological survival and reproduction processes in nature with continuous evolution through heredity, mutation, and natural selection [109,110]. The variances of the EA are genetic algorithm (GA), differential evolution (DE), estimation of distribution algorithm (EDA), DNA computing, Gene Expression Programming (GEP), and memetic algorithm (MA). The SIO algorithms are inspired based on the reproductive behaviour of the natural animal group and their hunting strategy [111]. The variances of swarm intelligence optimization algorithms are particle swarm optimization (PSO), ant colony optimization (ACO), cultural algorithm (CA), and bees algorithms, among others. Physics-based optimization algorithms are based on physics laws; the variance of these algorithms are the gravitational search algorithm (GSA), global dynamic harmony search algorithm (GDHS), and Jaya algorithm.

5.1. Modified metaheuristic methods

The applications of metaheuristic methods such as optimization techniques have greatly improved the accuracy and efficiency of AI in every field. However, metaheuristic algorithms have drawbacks, such as a slow convergence rate and local minimum problem that reduces applicability [111]. Studies seek to overcome these drawbacks by improving or modifying the existing optimization algorithms. The results of the improved optimization algorithms have proven their efficacies over the traditional optimization methods. Ref. [23] proposed an enhanced Al-Biruni Earth Radius optimization strategy using GA for feature selection. The procedure was applied to a stacked ensemble of learners; the base learners are LSTM and BiLSTM, and the meta learner is Hermite Neural Network for predicting wind speed and solar radiation. The prediction results revealed that the proposed model achieved less RMSE, MAE, MBE, and RRMSE than some existing methods. The method improved the prediction of RE. Adaptive opposition-based learning was enhanced with an artificial hummingbird algorithm (AHA) for developing an accurate solar cell system [112]. The adaptive opposition artificial hummingbird algorithm (AOAHA) was evaluated with 23 benchmark functions using the RMSE and integral absolute error (IAE). The results showed that the AOAHA performed better with the compared models.

Also, an improved Cuckoo search algorithm was applied to improve the forecasting accuracy of wind energy hybrid Least Square Support Vector Machine (LS-SVM) [113]. The accuracy and MAPE values revealed that the model is better than the existing ones. Table 3 shows

Table 3
Improved metaheuristic methods applied for RES

Reference	AI Model	Horizon	Data Site	Metaheuristic Algorithm	Improved Method	Application Area	Comparison Algorithm	Performance Metric	Findings
[114]	Type-2 Fuzzy NN	Daily	Denmark	Pruning Fuzzy Rules	Nonlinear Consequent Part Recurrent (NCPR)	Solar and Wind Systems	Type-2 Fuzzy System	RMSE	The application of evolutionary optimization will increase the convergence rate and accuracy.
[115]	ENN	Hourly	Hadoop	MFO	Big Data Analytic (BDA)	Solar Radiation	Persistence, LR, BPNN, LSTM, GRU, RNN, CSers	MSE RMSE	BDA contributed to a better improvement of the proposed method. However, implementing an improved metaheuristic method will overcome local minimum and easy convergence challenges.
[23]	Ensemble Bi-LSTM-hommer	Hourly	NASA	GA	Al-Biruni Earth Radius	Solar Radiation	PSO, GWO, WOA, GA, BER	r R ² RMSE MBE MAE RRMSE NSE WI	The proposed model outperformed the compared model based on superiority and generalization. The proposed model's validation with other databases is required to prove the generalization further.
[116]	FF-ANN	5mins	NREL Florida	SFLA	Gradient descent-SFLA (BP_SFLA)	Solar Radiation	SFLA-ANN, GA-ANN, BP-ANN	MAPE	Data preprocessing and implementation of deep learning will produce better performance.
[43]	MLP	30mins, 1, 3 and 6 Hours	Brazil	PSO	Ageing and Weaking Factors	Wind	BP, DE, PSO	MSE RMSE	The method searched for the best-hidden neurons and some logs and performed better.
[57]	HMD-IPSO-LSTM-ARMA-E	10mins	Changma China and Sotavento Spain	PSO	Learning factors change linearly with evolution	Wind	LSTM, RBF, Elman, BP	MSE RMSE MAE	More processing of wind speed attributes with the implementation of a more improved metaheuristic method will increase the performance of the proposed method.
[117]	FAC and SAC	Daily	Xinjiang China	CSA	Additive and Multiplicative Factors	Wind	BPNN, ARMA	RMSE MAPE	The adaptive nature of the method increases its performance; however, improving data quality will boost the accuracy of the forecasting model.
[118]	LSTM	5mins	Dingbian China	PSO	CM-LSTM	Wind power forecasting	IVMD-SE-Elman, MCC-LSTM, DBN, Elman	RMSE MAE TIC	Reconstruction, decomposition of wind data and replacement of the loss function of the LSTM improved the model's performance. However, an improved optimization algorithm will increase model performance.
[119]	SVM	Hourly	La Haute France	DA	Adaptive Learning Factor and DE Strategy	Wind power forecasting	BPNN, GPR, DA-SVM, GA-SVM, Grid-SVM	R ² RMSE MAE MAPE	The data decomposition method will improve model superiority.
[113]	LSSVM	10mins	China	CSA, PSO, GA, DE	Steepest Descent, Advanced PSO	Wind Speed	BPNN, Wavelet NN	MAE MRE RMSE MAPE IA Pbias	The model has better forecasting accuracy than the existing method; however, it requires more investigation based on data decomposition and implementation of deep learning.

the improved methods applied and their findings.

6. Hybrid AI in RES

In recent years, hybrid methods of AI have been studied and applied

in many areas of RES to address the shortcomings of single methods and to achieve better performance compared with single methods. Most of the time, in hybrid methods, one AI technique is used to optimize the decision boundary of the other [15]. For example, neuro-genetics is a combination of ANN and genetic algorithm; the latter is used for the

connection optimization of the previous. A typical conceptual framework of the hybrid application of AI is represented in Fig. 3. Table 4 summarizes the hybrid methods and the solar and wind RES findings.

In the study of the prediction of global solar radiation using a long-term meteorological dataset, [16] combined a support vector machine (SVM) with a firefly algorithm (FFA) to select the SVM parameters optimally. The results of the work were compared with the method of ANN optimization and genetic programming (GP) using performance metric MAPE, determination and correlation coefficient, and RMSE. The comparison revealed that SVM with FFA outperformed the ANN with GP. In [45], nine different optimization methods, ABC, ACO, BBO, ES, GA, GWO, PBIL, PSO, and TSA, were applied to feedforward MLP neural networks to predict wind energy. The study results proved that the GWO-MLP performed best based on computation time, MSE, mean training error rate (MTER) and Friedman's test values. Ref. [54] combined MLP NN with the Electromagnetic Field Optimization (EFO) algorithm; the proposed model was compared with the shuffled frog leaping algorithm (SFLA) and shuffled complex evolution (SCE). The RMSE, MAE and R2 results showed that EFO exhibited a fast convergence rate. Therefore, many optimization methods have been combined with conventional AI methods to optimally select the best parameters of the conventional AI and reduce their computation complexity. Also, a hybrid PSO-ANN strategy was developed to improve the accuracy of PV power prediction and to track the maximum global energy [130]. PSO-MLP and POS-ANFIS were applied for integrating the geothermal and solar systems in Ref. [51].

The applications of AI technologies have contributed immensely to the design of RES. A deep convolutional NN model called the SolarNet model for solar radiation forecasting was developed by [131]. This model applied rectified linear unit (ReLU) as an activation function of the convolution and output layers to decrease the risk of gradient vanishing. Comparing the SolarNet method using MAE with MLP, DT, SVM and LSTM, SolarNet performed better. Also, to predict future solar radiation power production, Ref. [132] combined knime software with deep learning methods. These deep learning models showed the feasibility of better prediction accuracy compared with conventional methods. In the quest to optimize and search for the maximum use of the energy produced by RES, some studies have explored reinforcement learning. Optimal control of RES using reinforcement learning was conducted, which provided higher accuracy than conventional AI [133,134]. Admas method and Keras-tuner optimization algorithms were combined with ANN to forecast solar power generation [68]. The accuracy of the results showed that the methods performed better than the single methods.

7. Findings and discussions

The demand for electrical energy is increasing exponentially due to the increase in industrialization and urbanization across the globe.

Research has established that the world can no longer depend on fossil fuel-based electrical energy generation due to greenhouse emissions, global warming, and other hazardous effects of the generation system. Undoubtedly, RES is the most promising human-friendly energy system, and its prospect has captivated the entire globe. The global investment in RES revealed that the system will take over the electric power generation system by over 50 % in 2050 [135,136]. Solar and wind energy systems are among the most prominent RES. In the quest to refine the adoption of RES, several studies have been carried out. Some of these studies forecasted the available output energy of RES, conducted optimal sizing of the RES, and estimated the grid system of RES, to mention a few. Many of these studies employed AI, while some employed physic-based strategies.

This study investigates AI's application in forecasting solar and wind renewable energy variability. As presented in Table 2, most studies implemented AI models to forecast the short-term horizon of solar and wind energy systems. Some of these studies performed data pre-processing to enhance the performance accuracy of the model. The performance metrics used in the studies revealed that the applied traditional models outperformed the adopted benchmarked models. It is important to note that the accuracy of the methods is based on their best capabilities. The strengths of these methods differ; generally, they can model nonlinear systems, parallel processing, flexibility in kernel selection, robustness to outliers, handling long-term dependencies, learning hierarchical representation, and handling variable-length sequences, to mention a few. The drawbacks of traditional algorithms are computational complexity, difficulties in hyperparameter tuning, overfitting, and lack of generalization. In order to enhance the forecasting accuracy of solar and wind systems, the hybrid strategy was implemented in some studies to address the drawbacks of traditional ML, as summarized in Tables 3 and 4.

Another driving force of the hybrid system is the "no free lunch theorem" of the ML algorithms because no algorithm has it all. Also, parts of the success achieved by hybrid methods are enhanced robustness, solutions to complex problems, adaptability and flexibility. However, hybrid methods are challenged with computational overhead, increased complexity and parameter tuning, and probable overfitting. Tables 3 and 4 show that the hybrid methods are in two forms. Some combinations of traditional methods like SVR-MLP [120], LSTM-CNN [20], and Fuzzy-SVR [63] were investigated.

Moreover, researchers have implemented hybridization of traditional methods and optimization algorithms such as GA, PSO, and GWO, to mention but a few. However, optimization algorithms have peculiar shortcomings, such as local minimum or premature convergence and poor search capability due to their internal weakness [13]. These shortcomings make it difficult for algorithms to perform trade-offs between exploration and exploitation in complex optimization problems [137]. Little effort has been contributed to improving the diversity of optimization algorithms, which is the emerging trend in AI technology.

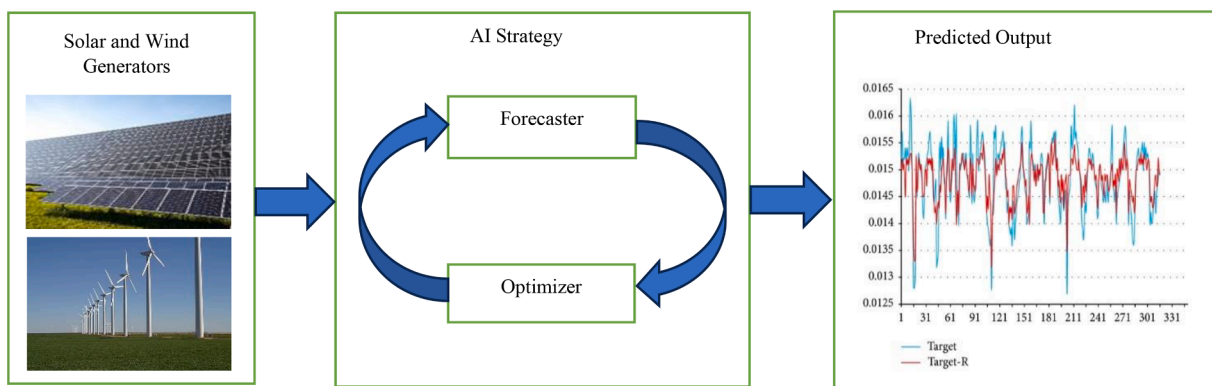


Fig. 3. A conceptual framework of hybrid AI for RES

Table 4
Hybrid Methods for RES

Reference	AI Model	Horizon	Data Site	Metaheuristic Algorithm	Application Area	Comparison Algorithm	Performance Metric	Findings
[120]	SVR-MLP	Daily	Iran	CSA	Solar Radiation	SVM, MLP	RMSE RRMSE MAPE	CSA selected the best parameters and weights for SVM; the proposed method proved superior over the compared methods. Modification of the CSA will increase the model's superiority. The proposed method reduced inappropriate dataset selection and hyperparameter estimation. Improving the optimization algorithm will enhance the model's accuracy.
[121]	Advanced-Mode ANN	Daily	Korea	PSO, Jaya Algorithm, FTMA	Solar and Wind	ARIMAX, Gradient Descent	R MSE RMSE MAE SDE BIC	
[122]	ELM	10mins, 1Hour	NTU and Lucia, Singapore	PSO	Solar power	LR, Auto-Regression	RMSE MAE	
[17]	LS-SVR	Monthly	Taiwan	GA	Solar power forecasting	ARIMA, SARIMA, GRNN, LS-SVR	RMSE MAPE	
[75]	SVM	Hourly	Victoria Australia	GA	Solar Radiation	SVM	RMSE MAPE	Implementing preprocessing and improved optimization methods will boost model superiority. Time series data decomposition using seasonal decomposition strengthened the model notwithstanding, an improved GA will impact the model's superiority. Exploring other weather parameters and data preprocessing will improve model accuracy. Also, modifying GA to keep the best population in the search space will enhance the performance of the proposed model. Implementation of the data preprocessing and optimization algorithm will overcome the challenge of easy convergence of the proposed model.
[63]	Fuzzy-SVM	Hourly	Turkey	-	Solar Radiation	ANFIS, Kernel Function	RMSE MAE MaxAE IQR-AE	
[123]	RNN	Hourly	Limpopo South Africa	GA	Solar Radiation	KNN	RMSE RRMSE MAE RMAE	
[16]	SVM	Monthly	Iseyin, Maiduguri, Jos Nigeria	FFA	Solar Radiation	ANN, GP	r R RMSE MAPE	
[54]	ANN	5mins	NASA	EFO	Solar Radiation	SCE, SFLA	R RMSE MAE	Good prediction accuracy: however, data preprocessing and improved FFA will enhance the prediction accuracy Application of the preprocessing method and modification of EFO will improve the proposed strategy. Improvement in feature selection and implementation of wind speed decomposition is required.
[124]	ANN	Hourly, Daily	United State	GA	Solar Radiation	Persistence, Perez model	R ² rRMSE RMSE MBE	
[125]	HMM-SVM	5, 15, 30mins	Bureau of Meteorology Australia	-	Solar Radiation	HMM, SVM	Accuracy	
[20]	LSTM-CNN	Hourly	Califonia USA	-	Solar Radiation	Smart Persistence, SVM, ANN, LSTM, CNN	R FS RMSE MAE	
[126]	LSTM	10mins	NREL China	Bayesian Optimization	Solar Radiation	Persistence, AR, SVR	nRMSE R FS	The proposed model requires an optimization method to address the local minimum challenge. Easy convergence threatens the proposed model; improving optimization will address the challenge. Cloud Detection demands investigation; also, improvement of BO will present a superior performance. It produced a significant performance compared with the existing models. Implementing other metaheuristic algorithms and other enhanced versions will improve the accuracy of the proposed method.
[42]	CNN	Hourly	Zuoying, Taiwan	Stochastic Optimization	Wind	SVM, RF, DT, MLP	RMSE MAE	
[127]	BPNN	Hourly	Lake Turkana	GA	Wind	-	MAE MSE RMSE	
[86]	ANN-RBF	Hourly	Iran	GA, PSO	Wind	ANFIS, ANN	MSE RMSE	
[41]	LSTM	Hourly	Spain and USA	SPSA, DE	Wind	BPNN, SVR, Persistence, DE-LSTM	R ² MAE MSE RMSE MAPE	2-stages wind energy data set preprocessing strategy improved the data quality. Still, there is room for improvement through enhanced

(continued on next page)

Table 4 (continued)

Reference	AI Model	Horizon	Data Site	Metaheuristic Algorithm	Application Area	Comparison Algorithm	Performance Metric	Findings
[128]	ELM	Hourly and Daily	Spain	CRO-SL	Solar, Wind and marine	CRO-ELM, CRO-MLR, SVR, MLP	R ² RMSE MAE	-SPSA or any other enhanced metaheuristic algorithm. Implementing feature extraction contributed to the performance of the proposed model; meanwhile, applying deep learning and auto-encoder will enhance the model.
[129]	BPNN	Hourly	Beijing	GA	Wind	BPNN, Persistence	RMSE MAE MAPE	Wind speed decomposition and improvement on GA will yield a better performance of the proposed model.
[19]	LSTM	Hourly	France	HBO	Wind power forecasting	LSTM, SVM, NN, GRACH, GA, PSO, DE, SSA, GWO, SCA	MAE MAPE RMSE MASE R ²	Low convergence speed and optimal solution problems of the model need investigation.
[67]	Dual-Stage Attention LSTM	Daily	North China	-	Wind power forecasting	XGBoost, MLP, LSTM, CNN-LSTM, TPA-LSTM, Transformer	MAE RMSE R	Implementation of data decomposition and metaheuristic algorithms will result in better performance.
[45]	ANN-MLP	Hourly	India	GWO	Wind Speed	GA, ABC, BBO, ES, PBIL, PSO, TSA,	MSE	Implementation of data preprocessing will improve the prediction accuracy of the proposed methods.
[44]	BPNN	10mins	China	GA	Wind speed	BPNN, GA-BPNN, HC-GA-BPNN, HC-EMD-GA-BPNN	MAE RMSE TIC	Integration of HC and VMD for data preprocessing enhanced model performance; conversely, an improved GA will produce a better performance

There are many fallow grounds to break regarding the potential of AI application in RES. Data cleaning and reprocessing are essential to boosting the accuracy of ML, DL and EL methods. Filtering, normalization [119], WPT [113], VMD [67], and more methods were applied for data preprocessing; still, this area requires more investigation to enhance the data quality for better prediction accuracy. The combination of traditional AI methods is scarce in RES forecasting; parallel combination and ensemble learning methods are robust strategies capable of addressing the shortcomings of traditional AI methods earlier mentioned. Optimization algorithms mainly focus on local search; they do not perform global exploration in the entire solution search domain. Implementing hybrid and improving optimization algorithms will bring diversity and balance the trade-off between exploration and exploitation to enhance the global exploration of the search domain.

8. Conclusion

The world expects RES to take over the electrical power industry soon to decarbonize global electrical generation. In achieving this task, the AI model has several roles to play. Many studies have implemented the AI model to prove its efficacy in forecasting the variability of solar and wind RES. This paper surveyed the existing studies on AI methods explored to forecast the variability of solar and wind energy systems. The taxonomy of different AI methods and conceptual framework of AI for predicting RES were presented. The forecasting methods and comparisons adopted by the surveyed articles were provided. Insights from each article surveyed were presented. Discussions and future directions based on the papers were presented. It was discovered that ML and its hybrid methods are receiving great attention in predicting the variability of solar and wind RES. However, many challenges are preventing the optimum performance of these AI models. These challenges and the future research recommendation based on the review findings are as follows:

- AI models such as ML and DL mainly depend on excellent and large datasets. Solar and wind energy data are very scarce due to the equipment cost for dataset collection. The lack of regional datasets makes regional forecasts of these renewable energies difficult. Most existing studies used datasets collected from a

single location or site. This makes the generalization of solar and wind renewable energy for a region a great challenge. Therefore, the availability of a sizeable regional dataset of solar and wind RES presents an excellent potential for better performance accuracy of some of ML and DL and implementation of big data analytics for forecasting of the RES. Big data analytics can handle large datasets and deduce valuable insight for better forecasting.

- The quest to increase the accuracy of RES forecasting models has led to combining several AI models. This, in turn, results in high computation costs. Implementing a metaheuristic model on the ML, DL, and EL can lead to higher computation time for the model. Modifying the metaheuristic model requires significant effort to reduce the forecasting models' computation time.
- The survey revealed that most studies forecasted very short-term and short-term horizons, few considered medium-term, and few studies were done on long-term horizons. There is a need to put more effort into the medium-term and long-term horizon to enable major maintenance and power system planning.
- Some existing solar and wind renewable energy forecasting models compromised their reliability due to the lack of real-time data acquisition and forecasting models. Implementation of the data assimilation method by integrating real-time observations and continuous updates of the model with a new set of data is required to enhance the forecasting accuracy of RES.
- Another challenge that is yet to be resolved in the solar and wind renewable energy forecast is spatiotemporal prediction. Temporal forecasting has received some attention; therefore, the focus must be shifted to studying the spatiotemporal variability of solar and wind renewable energy.
- Exploration of Blockchain Technology to create transparent, decentralized, and secure platforms for managing RES data will enhance the accuracy of RES forecasting.

CRedit authorship contribution statement

Blessing Olatunde Abisoye: Conceptualization, Resources, Methodology, Formal analysis, Writing – original draft. **Yanxia Sun:** Conceptualization, Supervision, Validation, Writing – review & editing, Funding acquisition, Investigation, Project administration, Resources.

Wang Zenghui: Conceptualization, Formal analysis, Funding acquisition, Investigation, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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